2421 is also self complementing code.

If summation of all the weights is equal to 9 then that code is self complementing code. thus 5211 is also self complementing.

Excess-3 Code Continued

Decimal $\longrightarrow$ 8421 BCD $\xrightarrow{+0011}$ Excess-3 Code.

$$(23)_{10} \rightarrow \begin{array}{r} 0010 \quad 0011 \\ +\ 0011 \quad 0011 \\ \hline 0101 \quad 0110 \end{array}$$

$(XX.XX)_b \longrightarrow (\ )_{10} \longrightarrow$ 8421 BCD $\xrightarrow{+0011}$ Excess-3code.

(d) Biquinary Code :-

* length of 1 digit = 7 bit.
* It is a weighted code.
* It is used for error detection.

Decimal number is represented in such a way that only 2bits are high and rest 5 bits are low.

5.

Decimal Digit	Biquinary	
	5 0 4 3 2 1 0	→ ~~0+0~~
0	0 1 0 0 0 0 1	→ ~~0+1~~ 0+0
1	0 1 0 0 0 1 0	→ 0+1
2	0 1 0 0 1 0 0	→ 0+2
3	0 1 0 1 0 0 0	→ ~~0+4~~ 0+3
4	0 1 1 0 0 0 0	→ 0+4
5	1 0 0 0 0 0 1	→ 5+0
6	1 0 0 0 0 1 0	→ 5+1
7	1 0 0 0 1 0 0	→ 5+2
8	1 0 0 1 0 0 0	→ 5+3
9	1 0 1 0 0 0 0	→ 5+4

Limitation :- It requires 7 bits, length of the datapackage becomes high as it requires double space.

GRAY CODES

* Non-weighted code.
* Cyclic code. → two successive nos. differ by one 1 bit.
* Unit distance code

4- bit Gray code

0 → 0000
1 → 0001
2 → 0011
3 → 0010
4 → 0110
5 → 0111
6 → 0101
7 → 0100
8 → 1100
9 → 1101
10 → 1111
11 → 1110
12 → 1010
13 → 1011
14 → 1001
15 → 1000

Binary codes

0 → 0000

Gray codes are used in position encoder. and rotatory encoder.

BINARY TO GRAY

B_3	B_2	B_1	B_0	G_3	G_2	G_1	G_0
1	0	1	0	1	1	1	1

MSB of Binary & Gray will be same

$G_3 = B_3 = 1$

$G_2 = B_3 \oplus B_2 = 1 \oplus 0 = 1$

$G_1 = B_2 \oplus B_1 = 0 \oplus 1 = 1$

$G_0 = B_1 \oplus B_0 = 1 \oplus 0 = 1$

G_3	G_2	G_1	G_0
1	1	1	0

Binary

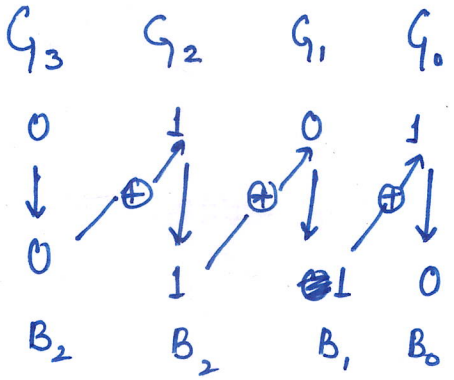
B_3 B_2 B_1 B_0

$B_3 = G_3 = 1$

$B_2 = B_3 \oplus G_2 = 1 \oplus 1 = 0$

$B_1 = B_2 \oplus G_1 = 0 \oplus 1 = 1$

$B_0 = B_1 \oplus G_0 = 1 \oplus 0 = 1$



Q find Binary equivalent of the given gray codes.

(1) 1001101

(2) 1100110

BCD Addition

Case-1:- Sum equals 9 or less with carry 0.

3 + 6

```
  0 1 1 0
  0 0 1 1
-------------
0| 1 0 0 1
```

carry.

5 + 2

```
  0 1 0 1
+ 0 0 1 0
-------------
0| 0 1 1 1
```

↓ carry.

The addition is carried out as normal binary addⁿ

Case-2:- Sum greater than 9 with carry 0.

6 + 8

~~0 1 1 0~~
~~0 1 0~~

```
  0 1 1 0
+ 1 0 0 0
-------------
0 1 1 1 0  > 9.
```

↳ invalid BCD

The sum is corrected by addition of six (0 1 1 0) to the invalid BCD.

```
  1 1 1 0
  0 1 1 0
-------------
1 0 1 0 0
```

6 + 8 = 1 4.

→ for Case-3

Note:- If a carry is produced from the first block, add it to next higher order BCD digit

Case-3 Sum equal 9 or less with carry 1.

$8 + 9 = 17$.

```
  1 0 0 0
  1 0 0 1
-------------
1) 0 0 0 1   (Incorrect result).
```

0001 0001.

To correct the result +6 is to be added to least significant digit sum.

```
      1 0 0 0
      1 0 0 1
-----------------
0 0 0 1 0 0 0 1
0 0 0 0 0 1 1 0
-----------------
0 0 0 1 0 1 1 1
-----------------
      1     7
```

Q. 24 + 18

48 + 58

Q. 96 + 56

Q. 954 + 463.

BCD Subtraction using 9's complement.

* Represent -ve nos. in 9's comp.

find 9's comp of $(3497)_{BCD}$.

$$\begin{array}{r} 9\ 9\ 9\ 9 \\ -\ 3\ 4\ 9\ 7 \\ \hline 6\ 5\ 0\ 2 \end{array}$$

9's comp of $(3497)_{BCD} = 6502$

- **Step-1** :- find 9's comp of negative no.

Step-2 :- Add two numbers using BCD add^n. (first no. & 9's comp of 2nd no)

- **Step-3** :- If carry is not generated result is -ve, find the 9's comp. of the result, otherwise result is +ve and add carry to the result.

Q $(46)_{10} - (22)_{10}$ in BCD using 9's comp.

find 9's comp of 22

$99 - 22 = 77$

Add 46 + 77

$$\begin{array}{r} 0100\ \ 0110 \\ 0111\ \ 0111 \\ \hline 1011\ \ 1101 \\ 0110\ \ 0110 \\ \hline 10010\ \ 0011 \\ \quad\quad\quad\quad +1 \\ \hline 0010\ \ 0100 \end{array}$$

add +6 to each digit

Ans $(24)_{10}$

Q. $(24)_{10} - (56)_{10}$.

9's comp of 56

$99 - 56 = 43$.

$$\begin{array}{r} 24 \\ +43 \\ \hline \end{array} \qquad \begin{array}{r} 0010\ 0100 \\ 0100\ 0011 \\ \hline 0110\ 0111 \end{array} \Rightarrow 67.$$

No carry is generated

Take 9's comp of 67

$= 99 - 67$

$= 32$

Ans. $(-32)_{10}$.

Q. Subtract 748 from 983 BCD No.

Take 9's comp of 748

$$\begin{array}{r} 999 \\ -748 \\ \hline 251 \end{array}$$

$$\begin{array}{r} 983 \\ -748 \\ \hline 235 \end{array}$$

983 ⇒	1001	1000	0011
251	0010	0101	0001
	1011	1101	0100
+	0110	0110	0000
	1) 0010	0011	0100
			+1
	0010	0011	0101

$(235)_{10}$.

BCD Subtraction using 10's comp

Step-1 :- find 10's comp. of -ve no.

Step-2 :- Add two nos. using BCD addn (first no & 10's comp. of 2nd)

Step-3 :- If carry is not generated result is -ve, find the 10's comp of result, otherwise result is +ve & discard carry.

Q $(46)_{10} - (22)_{10}$

find 10's comp. of 22

$(99-22)+1 = 78$

```
  46      0100  0110
+ 78      0111  1000
          ----------
          1011  1110   → Invalid BCD
        +  0110  0110  → add 6 to each digit
          ----------
        1) 0010 0100
          ----------
           2    4.
```

$(24)_{10}$

Q. $(24)_{10} - (56)_{10}$ in BCD using 10's complement.

10's comp of 56 = 9's comp + 1

= 99 - 56 + 1

= 44.

```
  0010 | 0100
  0100 | 0100
  ---------------
  0110 | 1000   (68)BCD
```

No carry is generated, result is -ve., take 10's comp of (68)

99 - 68 + 1 = 32 Ans $(-32)_{10}$

Error Detection & Correction

① Parity Bit

* A parity bit is used for the purpose of detecting errors during transmission of binary info.
* A parity bit is an extra bit included with the binary message to make the no. of 1's either even or odd.
* The message, including the parity bit is transmitted and then checked at the receiving end for errors.
* An error is detected if the checked parity does not correspond with the one transmitted.
* The circuit that generates the parity bit is the transmitter is called a parity generator.
* The circuit that checks the parity in the receiver is called parity checker.
* In even parity, the added parity bit will make the total no. of 1's an even no. and similarly, in odd parity, the added bit will make the total no. of 1's an odd no.

A	B	C	Msg with odd parity		Message with even parity	
0	0	0	000	1	000	0
0	0	1	001	0	001	1
0	1	0	010	0	010	1
0	1	1	011	0	011	0

Q. Write an ASCII code for the decimal digit 9 with an even parity. Place parity in MSB.

ASCII Code for 9 is 0111001

Added parity bit :- 00111001.

HAMMING CODE

* The no. of parity bits depends on the no. of data bits

If x is no. of data bits.
P is no. of parity bits.

$$2^P \geq x + P + 1$$

let suppose $x = 4$.

P is found by hit & trial.

let $P = 2$.

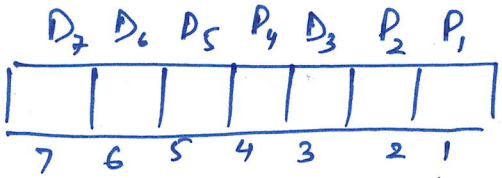
$$4 \geq 4 + 2 + 1$$
$$4 \geq 7 \times$$

$P = 3$.

$$8 \geq 4 + 3 + 1$$
$$8 \geq 8 \checkmark$$

thus $P = 3$.

No. of data bits = 4
No. of parity bits = 3
= 7



Parity bits are located at positions that no. 2^n ($n = 0, 1, \cdots m$)

Bit Designation

D_7	D_6	D_5	P_4	D_3	P_2	P_1
7	6	5	4	3	2	1
111	110	101	100	011	010	001.

Assignment of P_1

* P_1 has 1 for its rightmost digit.
* This parity bit checks all bit locations, including itself, that have 1's in the same location in the binary location nos.
* Thus P_1 checks → 1, 3, 5 and 7.

Assignment of P_2

P_2 has 1 for its middle digit

P_2 checks bit location 2, 3, 6, 7.

Assignment of P_4

P_4 has 1 for its leftmost bit

P_4 checks bit location 4, 5, 6, 7.

Q. ~~find the no. of parity bit req.~~

Q. Encode the binary word 1011 into seven bit even parity hamming code.

$2^P = 2^3 = 8$.

$x + P + 1 = 4 + 3 + 1 = 8$

No. of parity bit = 3.

D_7	D_6	D_5	P_4	D_3	P_2	P_1
1	0	1	0	1	0	1

$P_1 =$ D_3 D_5 D_7

(1) 1 1 1

$P_2 =$ D_3 D_6 D_7

(0) 1 0 1

$P_3 =$ D_5 D_6 D_7

(0) 1 0 1

So, hamming code (1 0 1 0 1 0 1)

Q Determine the single-error correcting code for the ~~info~~ data bits 1 0 1 1 1 for odd parity.

for $P = 3$ $2^3 = 8 \geq 5 + 3 + 1 =$

$8 \geq 9$ ✗

for $P = 4$ ✓ $16 \geq 5 + 4 + 1$

$16 \geq 10$

Total no. of bits = $5 + 4 = 9$

	D_9	P_8	D_7	D_6	D_5	P_4	D_3	P_2	P_1
	1001	1000	0111	0110	0101	0100	0011	0010	0001
Data bits	1	0	0	1	1	1	1	1	0

Odd parity

P_1 will check for 1, 3, 5, 7, 9

$P_1 = 0$

P_2 will check for 2, 3, 6, 7

$P_2 = 1$

P_4 will check for 4, 5, 6, 7

$P_4 = 1$

P_8 will check for 9

$P_8 = 0$

Hamming code :- 100111110.

Q. Determine Hamming code sequence with odd parity for natural BCD for making it an error correcting code.

	D_7	D_6	D_5	P_4	D_3	P_2	P_1
0000	0	0	0	1	0	1	1
0001	0	0	0	1	1	0	0
0010	0	0	1	0	0	1	0
0011	0	0	1	0	1	0	1
0100	0	1	0	0	0	0	1
0101	0	1	0	0	1	1	0
0110	0	1	1	1	0	0	0
0111	0	1	1	1	1	1	1
1000	1	0	0	0	0	0	0
1001	1	0	0	0	1	1	1